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Beam Angle Selection of Non-Coplanar Fields in Proton Therapy

Advancing automated treatment planning of cranial tumours

Master's thesis presentation :: September 20th 2023

Introduction

- Proton therapy and Fields
- Automated Treatment Planning

1) Convolutional Neural Networks (CNNs) for Single Non Coplanar Field

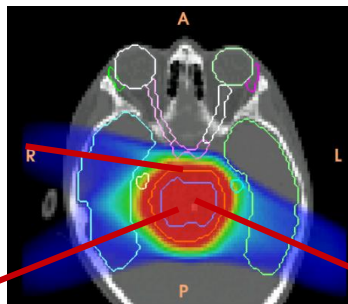
- Methods : CNNs, Dataset, Inputs
- Supervised learning
- Unsupervised learning

2) Three Fields Dose Descent (3FDD)

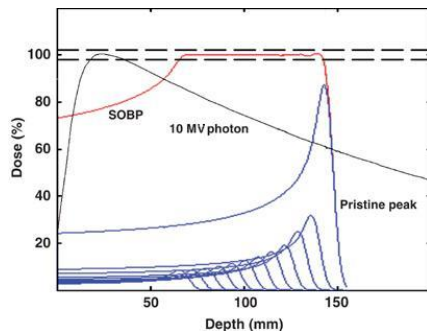
- Algorithm
- Results on convergence, plans & incidences

Conclusion & Outlook

Introduction



Dose distribution for three non coplanar fields

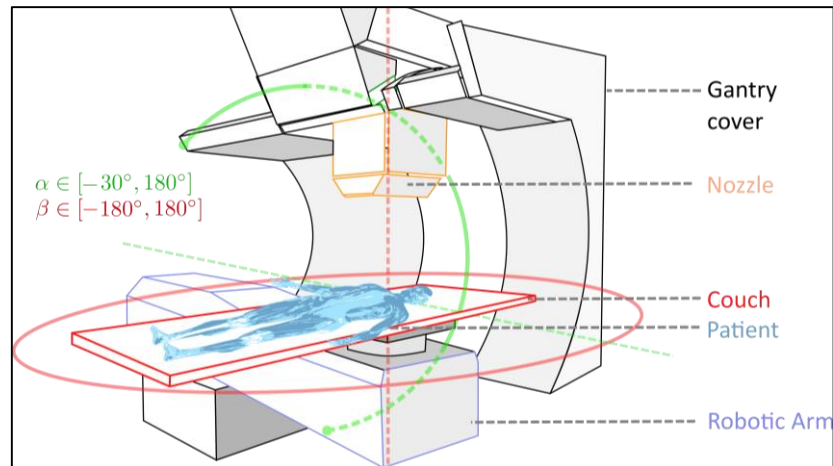


Spread Out Bragg Peak [1]

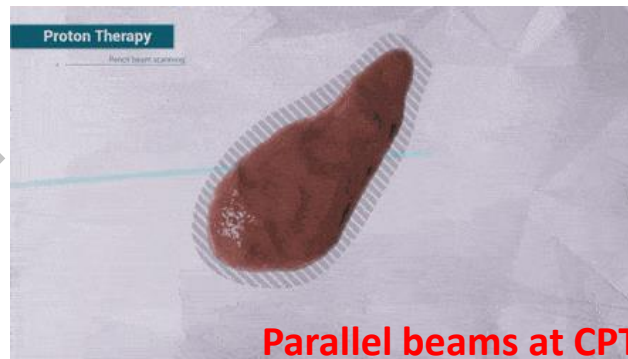
Field definition :

- Target structure
- Field Weight
- Gantry angle α
- Couch angle β

Define "Incidence"



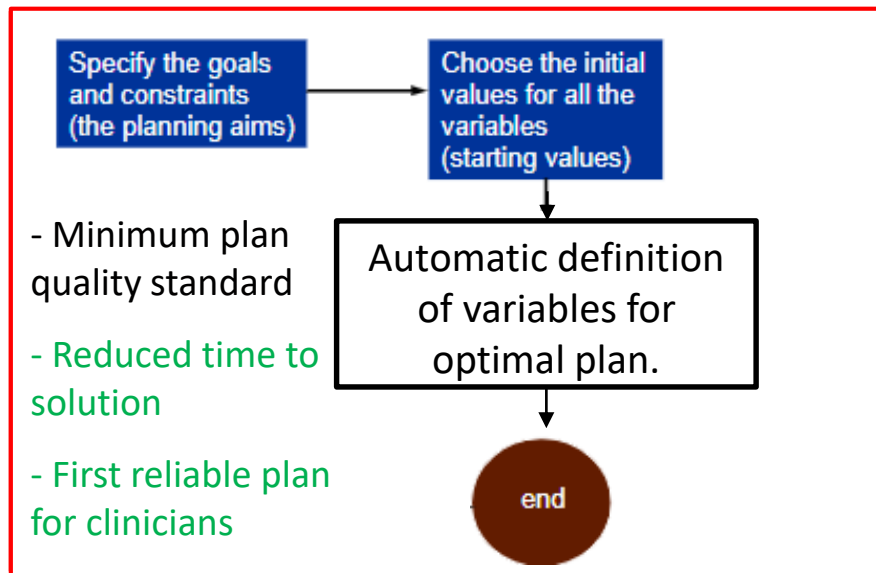
Treatment room of Gantry 2 at CPT.
Patient at $(\alpha, \beta) = (0, 90^\circ)$



Pencil Beam Scanning
in Proton therapy [2]

Treatment Planning

“Process by which a [proton-therapy] device is programmed to deliver an amount of protons to a patient.” [3]



Automated Planning Process

Goal : select optimal beam angles/incidence for non-coplanar fields automatically

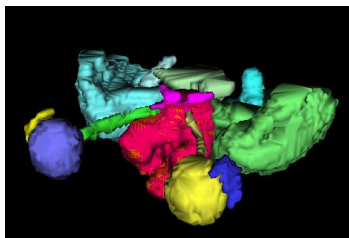
[3] Kevin L. Moore, Automated Radiotherapy Treatment Planning, Seminars in Radiation Oncology, Volume 29, Issue 3, 2019, Pages 209- 218, ISSN 1053-4296, <https://doi.org/10.1016/j.semradonc.2019.02.003>.

[4] Schneider, U., Lomax, A.J. , Modern Treatment Planning. 402-0343-00L Physics Against Cancer: The Physics of Imaging and Treating Cancer, ETH (2023)

Take into account

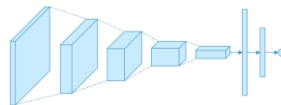
- Dosimetric prescriptions
- Patient structures

Structure Name	Constraint Quantity	Threshold	Bound kind
Brainstem	D_2	63 Gy	UPPER
CBS	D_2	54 Gy	UPPER
Chiasm	D_{mean}	54 Gy	UPPER
Chiasm	D_2	60 Gy	UPPER
Optic nerve left	D_2	60 Gy	UPPER
Optic nerve right	D_2	60 Gy	UPPER
Temporal lobe left	D_{max}	103.00%	UPPER
Temporal lobe right	D_{max}	103.00%	UPPER
...	...	...	...



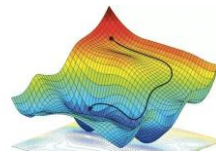
Two Novel Approaches

1) Convolutional Neural Networks



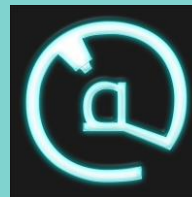
- supervised
- unsupervised

2) Modified Gradient Descent



New Framework

Automatic and Rapid tool for
Optimal Beam Angle Selection
(AROBASE, or @) in **julia**



Convolutional Neural Networks

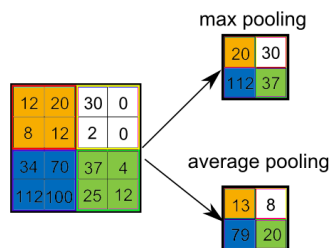
Angle prediction for a
single non-coplanar field

Convolution



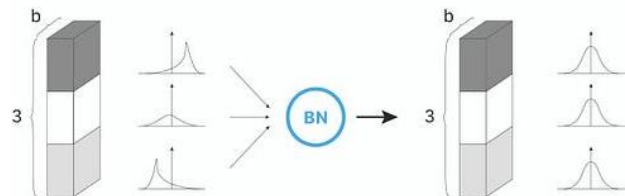
- Receptive fields
- Filters = learnable parameters

Pooling

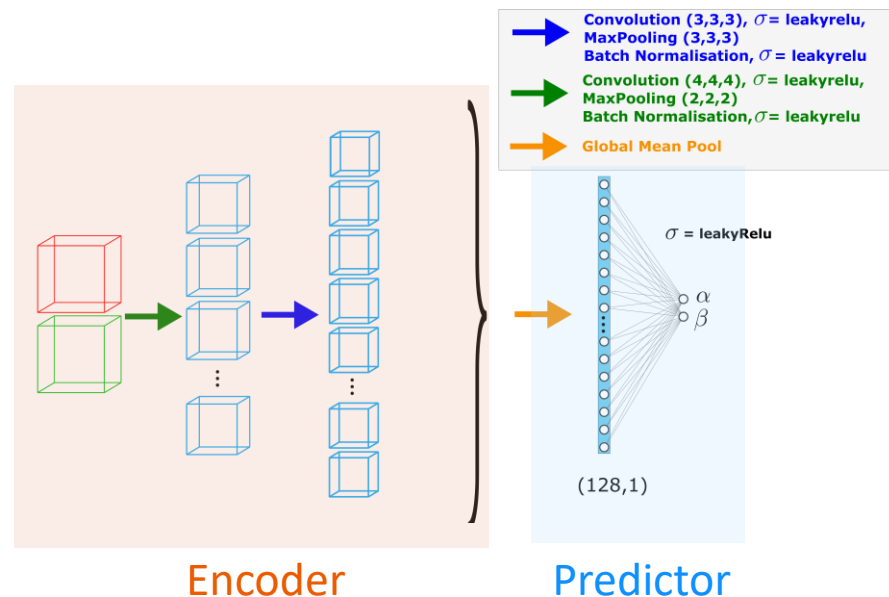


- Down sampling
- Diminish complexity

Batch Normalisation

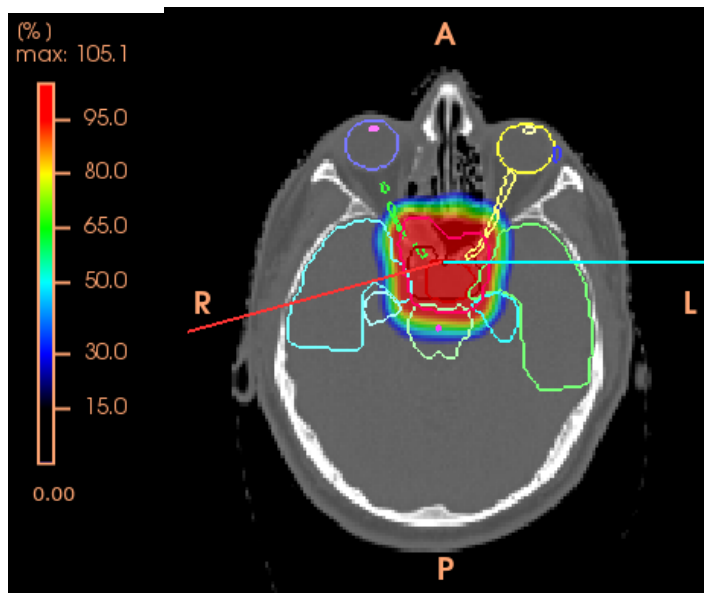


- Normalise channels independently
- Faster, more stable training



Dataset

- 94 **patients** : 75 for training, 19 for test
- 2013-2020, treated at CPT's Gantry 2
- Intra/Extra-cranial tumours (inhomogeneous)



For each **patient**

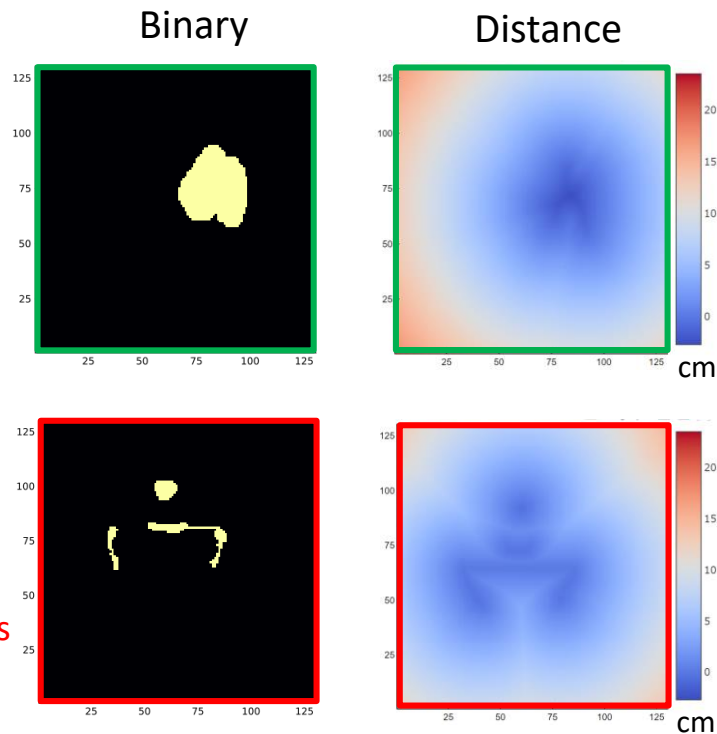
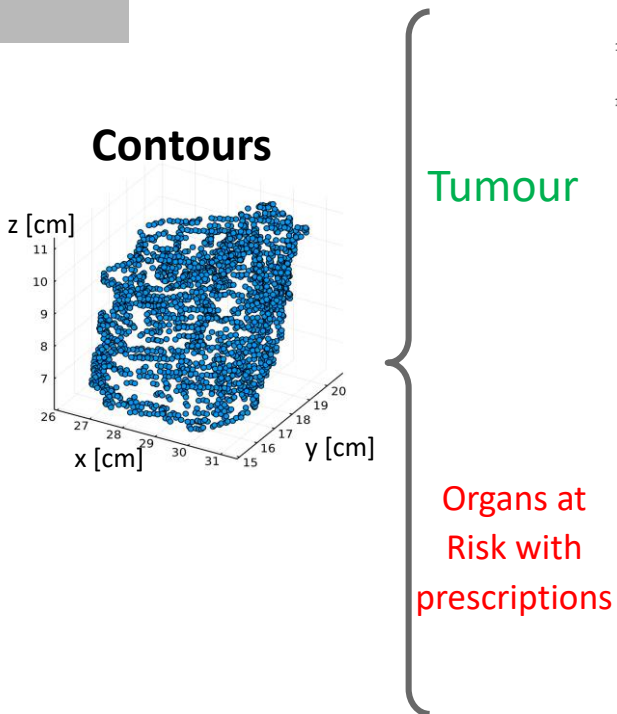
- Computed Tomography
- Contours of structures
- Reference dose distribution
- Reference Clinical Treatment Plan :
 1. Dosimetric prescriptions (= plan goals)
 2. Reference Fields (weights, angles (α^* , β^*), targets)

Closest to ground truth, but other possibilities!

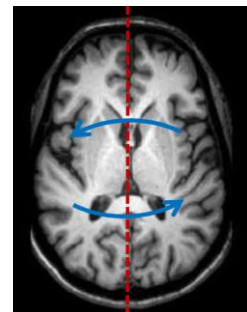
Methods - Input Masks

Problem: What inputs for the neural networks ?

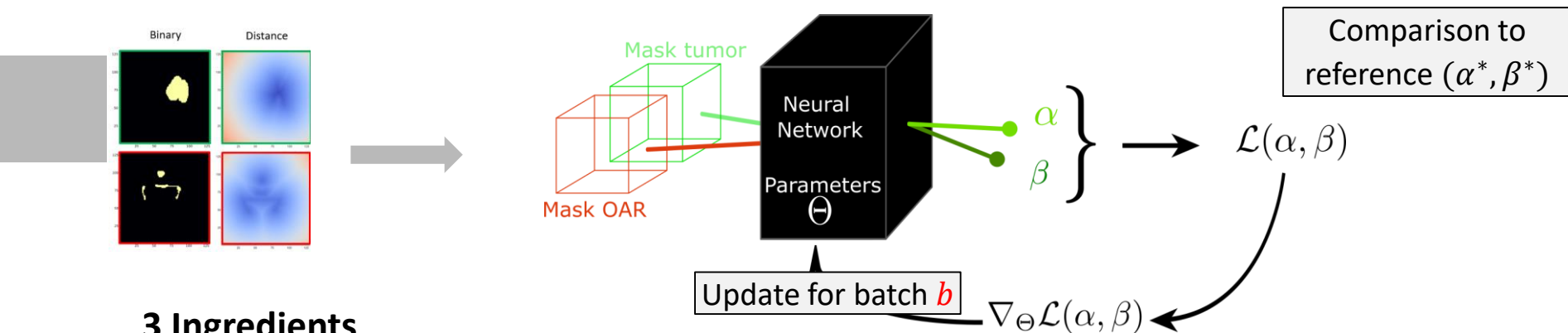
Voxel value =
In-slice distance to
closest structure



- Standardised
- Maximise information density
- Augmentation using mirror symmetry (x2 training set)



Supervised Approach



3 Ingredients

1) Optimiser: Adam [5], learning rate η

2) Loss function: $\mathcal{L}(\alpha, \beta) = \underbrace{\frac{1}{2} ((\alpha - \alpha^*)^2 + (\beta - \beta^*)^2)}_{\text{MSE}} + \underbrace{\lambda_{reg} L_{reg}}_{\text{Regularisation}}$

3) Networks

- “Full”
- “Reduced” (prevent overfitting)
- “Inception” (based on Inception Module [6])

$$\left\{ \begin{array}{l} L_1 = \sum_{\theta \in \Theta} |\theta| \\ L_2 = \sum_{\theta \in \Theta} \theta^2 \end{array} \right. \quad \text{To prevent overfitting}$$

[5] Diederik P. Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization. 2017. arXiv: 1412.6980[cs.LG]

[6] Christian Szegedy et al. “Going deeper with convolutions”. In: 2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). 2015, pp. 1–9. DOI: 10.1109/CVPR.2015.7298594.

Supervised - Best Experiments : Binary

Hyper-parameter grid search

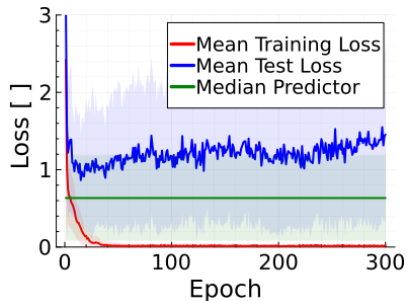
$b, \eta, \lambda_{reg}, L_{reg}$

Best := experiment with best mean test loss over all epochs

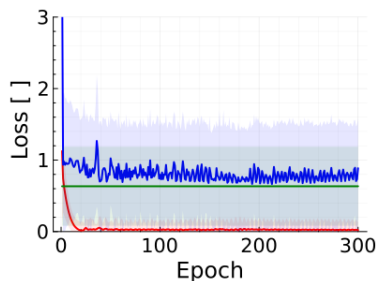
Median predictor

Median of training set is prediction of test patients

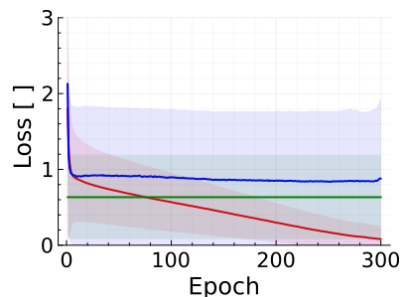
Full



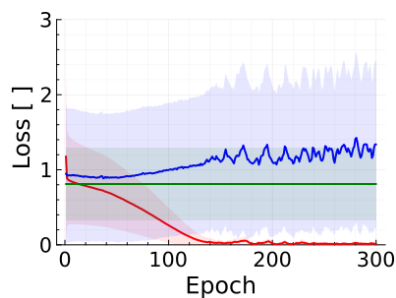
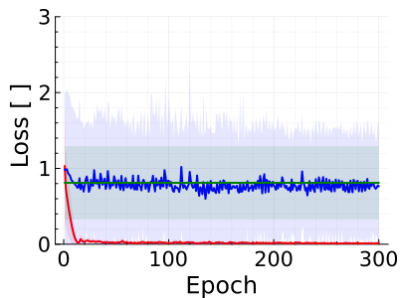
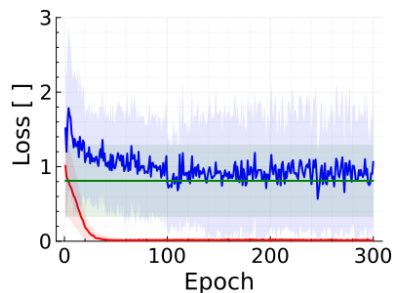
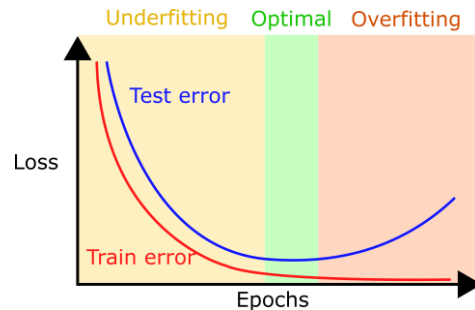
Inception



Reduced



Initial dataset



Augmented dataset

Supervised - Best Experiments : Distance

Hyper-parameter grid search

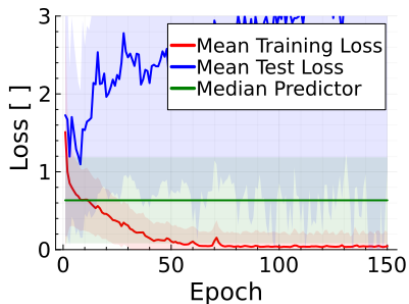
$b, \eta, \lambda_{reg}, L_{reg}$

Best := experiment with best mean test loss over all epochs

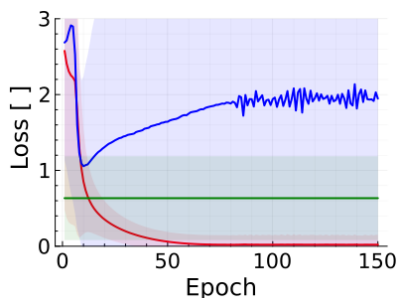
Median predictor

Median of training set is prediction of test patients

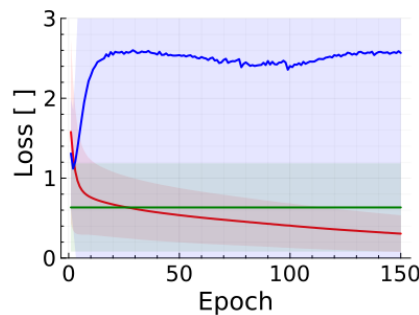
Full



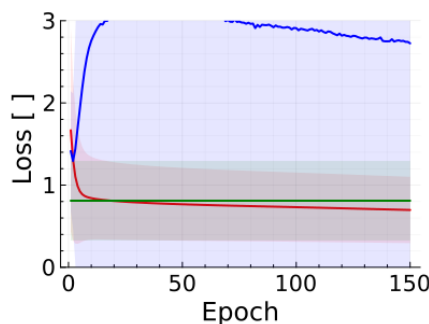
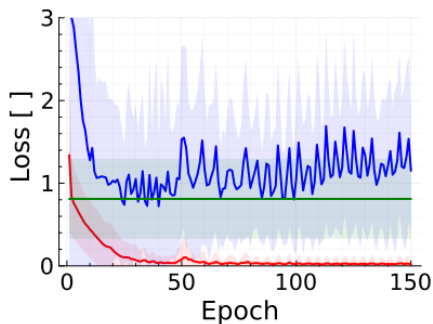
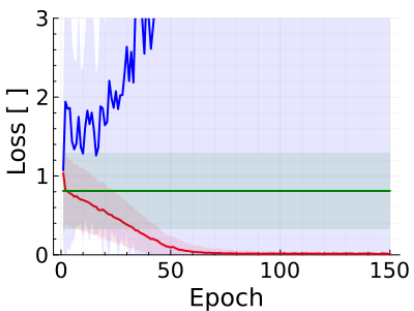
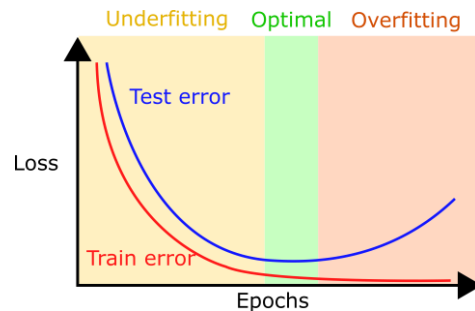
Inception



Reduced



Initial dataset

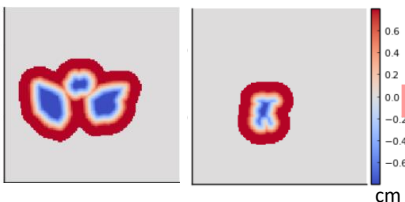
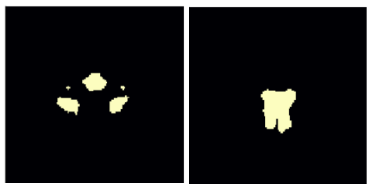


Augmented dataset

Supervised - Gradient Based Activation Maps

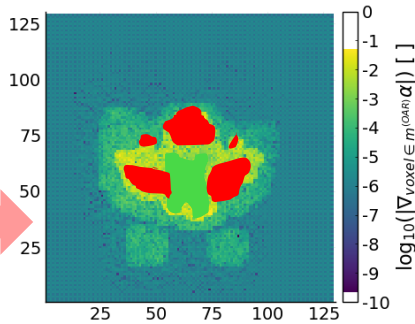
Do the networks perceive and distinguish the structures?

Inputs

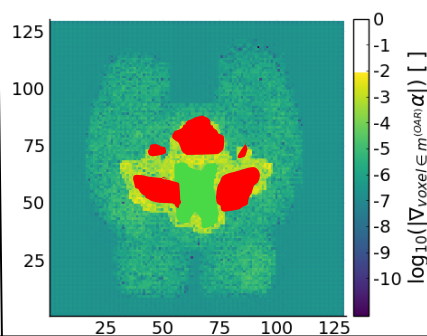


Variation
of output
w.r.t input

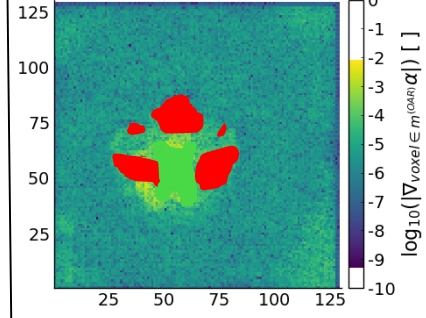
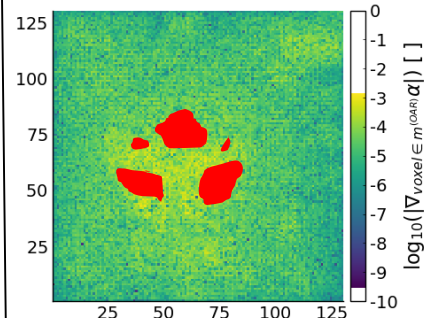
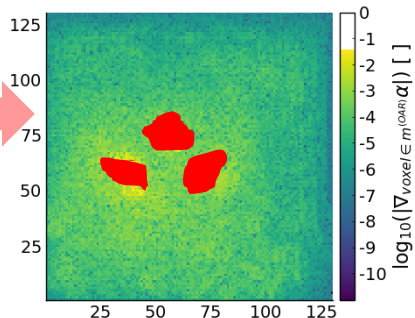
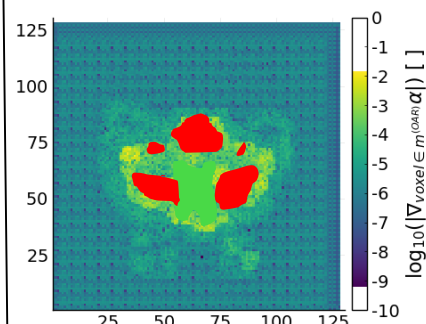
Full



Inception



Reduced

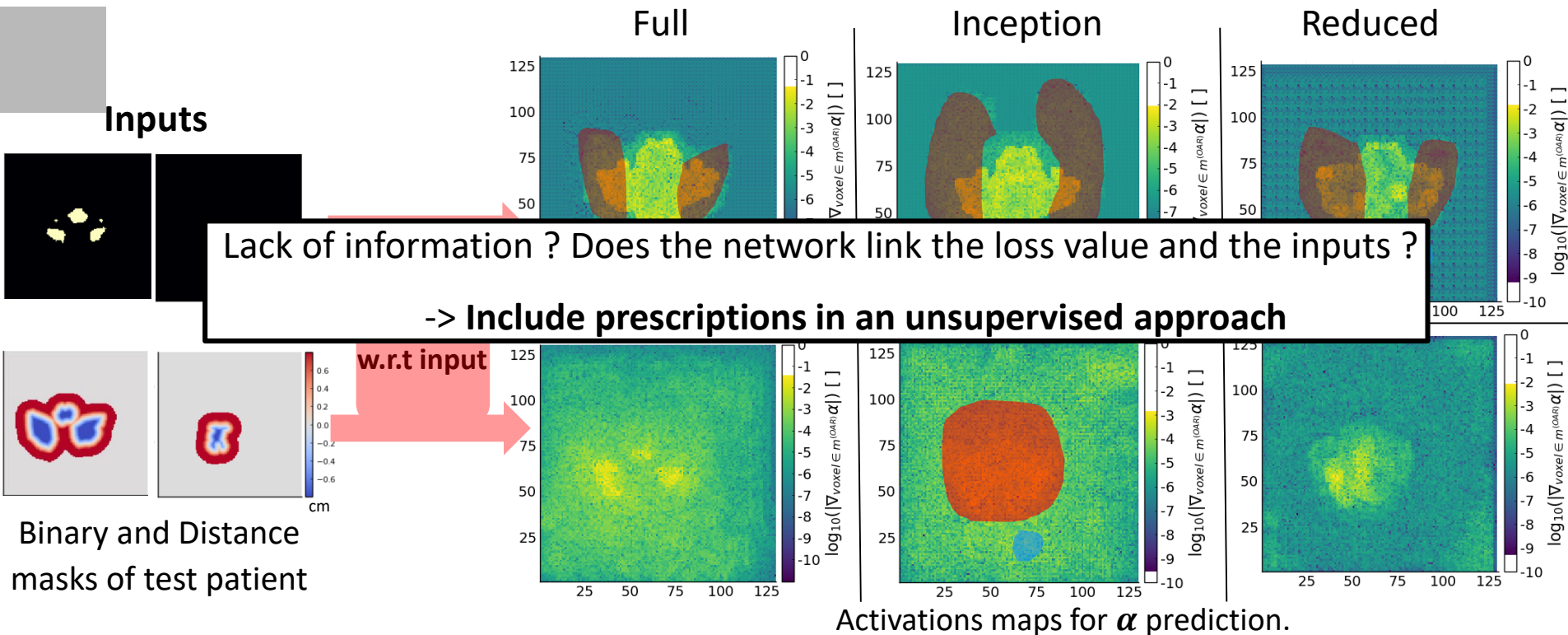


Activations maps for α prediction.

Gradient with respect to **Organs at risk** input mask.

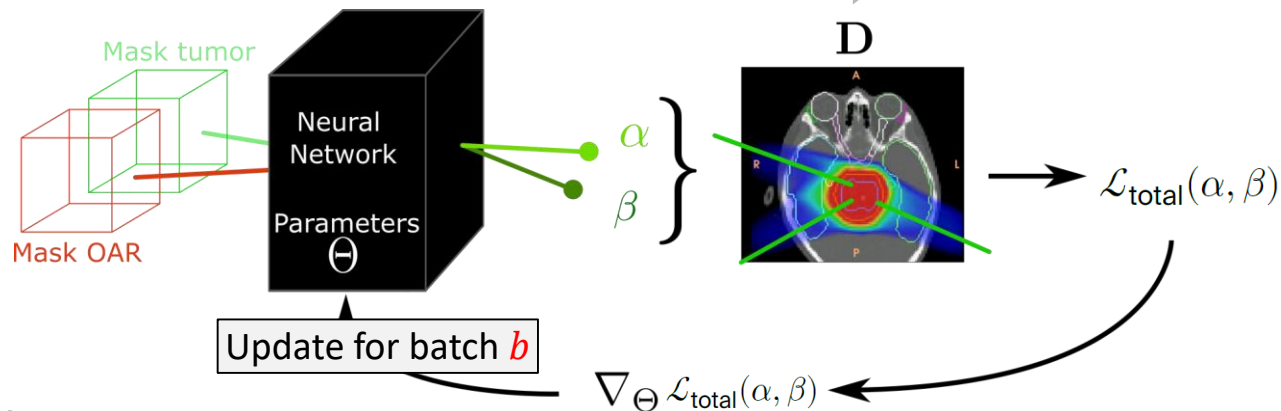
Supervised - Gradient Based Activation Maps

Do the networks perceive and distinguish the structures?



Gradient with respect to **Organs at risk** input mask.

Unsupervised Approach



3 Ingredients

1) Optimiser: Adam, learning rate η

2) Loss: $\mathcal{L}_{\text{total}}(\alpha, \beta) = [\mathcal{L}_{\text{D}}^* + \lambda_{\pi} L_{\pi} + \lambda_{\text{reg}} L_{\text{reg}}] (\alpha, \beta)$

Prescriptions compliance [7]

Physicality
of angles

L1 or L2 regularisation

3) Networks: “Full”, “Reduced”, “Inception”

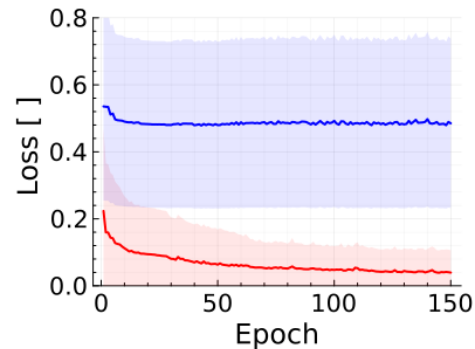
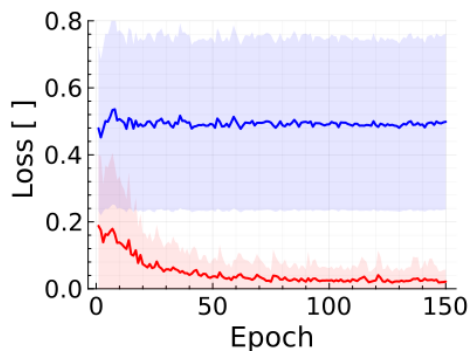
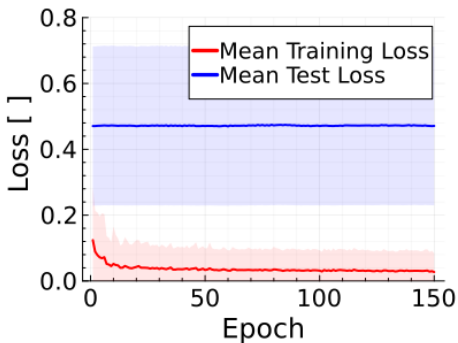
Unsupervised - Best Experiments

Hyper-parameter grid search

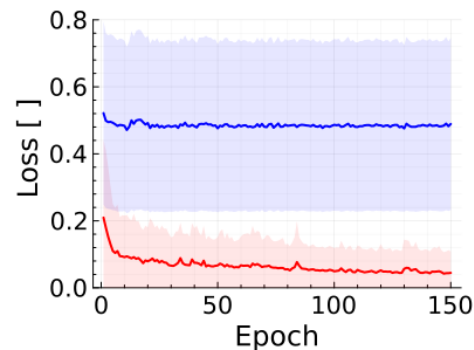
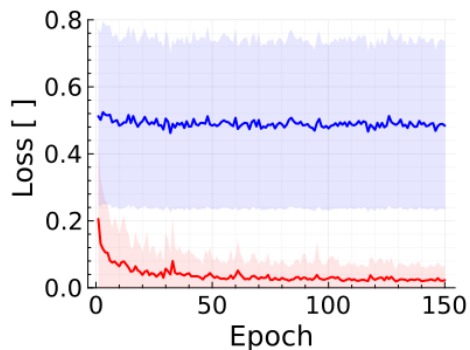
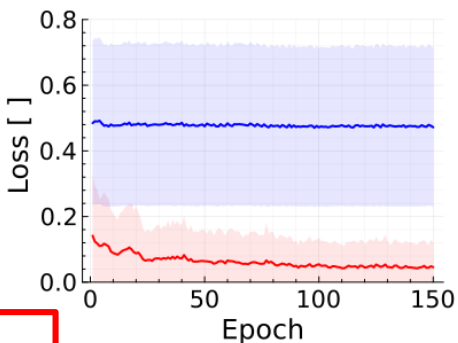
Best := experiment with best mean test loss over all epochs

$b, \eta, \lambda_{\pi}, \lambda_{reg}, L_{reg}$

Binary



Distance



**Losses
Normalised !**

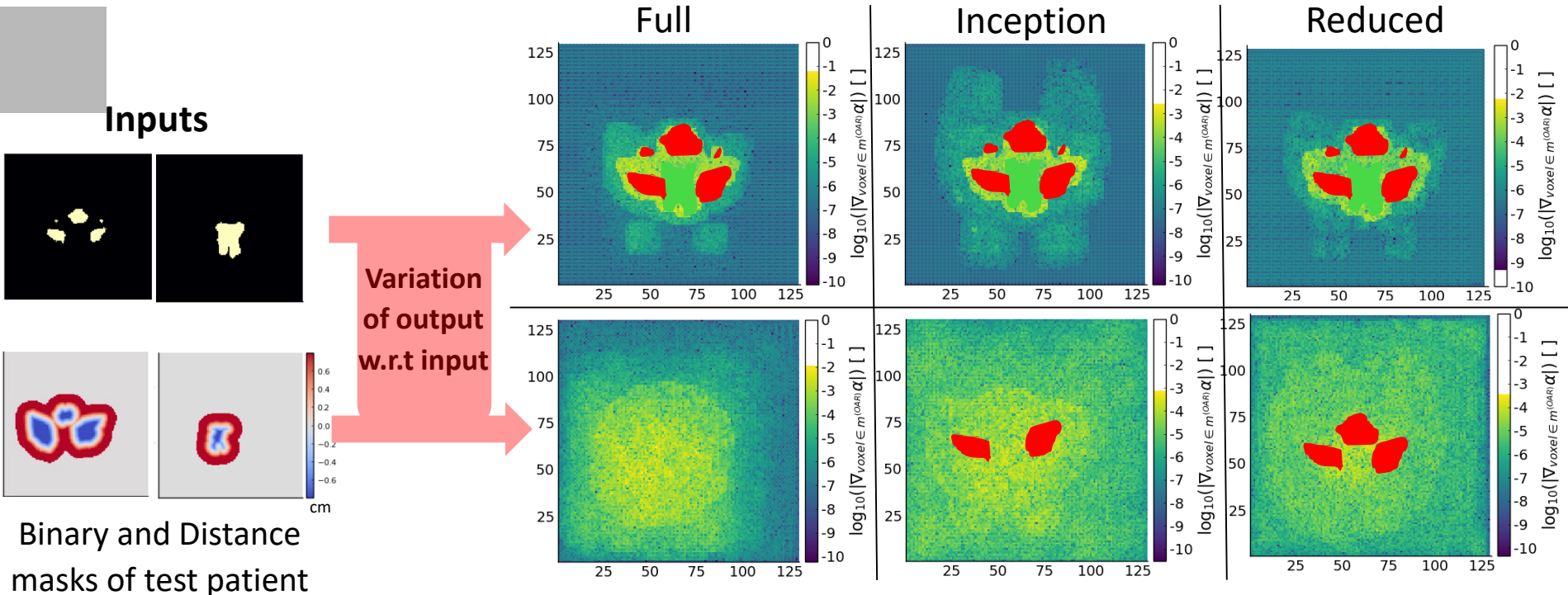
Full

Inception

Reduced

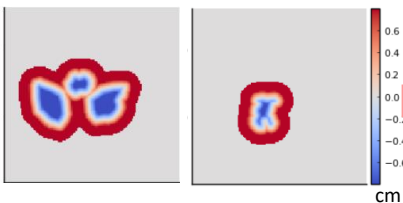
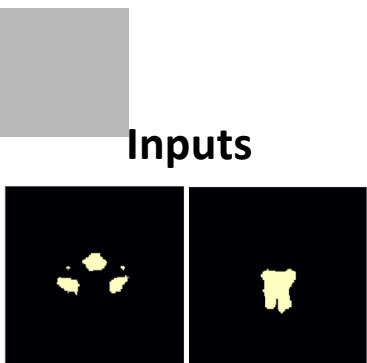
Unsupervised - Gradient Based Activation Maps

Do the networks perceive and distinguish the structures?

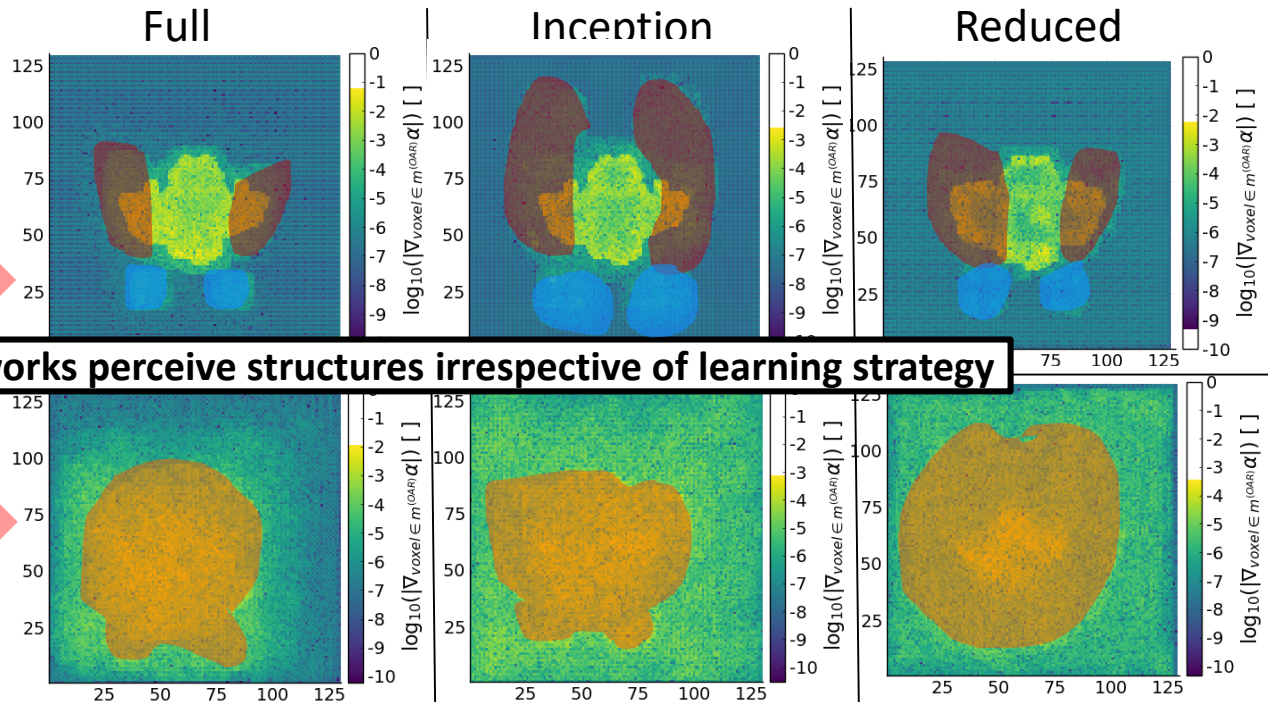


Unsupervised - Gradient Based Activation Maps

Do the networks perceive and distinguish the structures?



Variation of output w.r.t input



-> Networks perceive structures irrespective of learning strategy

Activations maps for α prediction. Gradient with respect to **OAR** mask

Supervised

- Models: learn, fail to generalise (overfit)
- Median predictor reached in some cases
- Better performance from Inception
- Perception of structures
- No conclusion for augmentation

Unsupervised

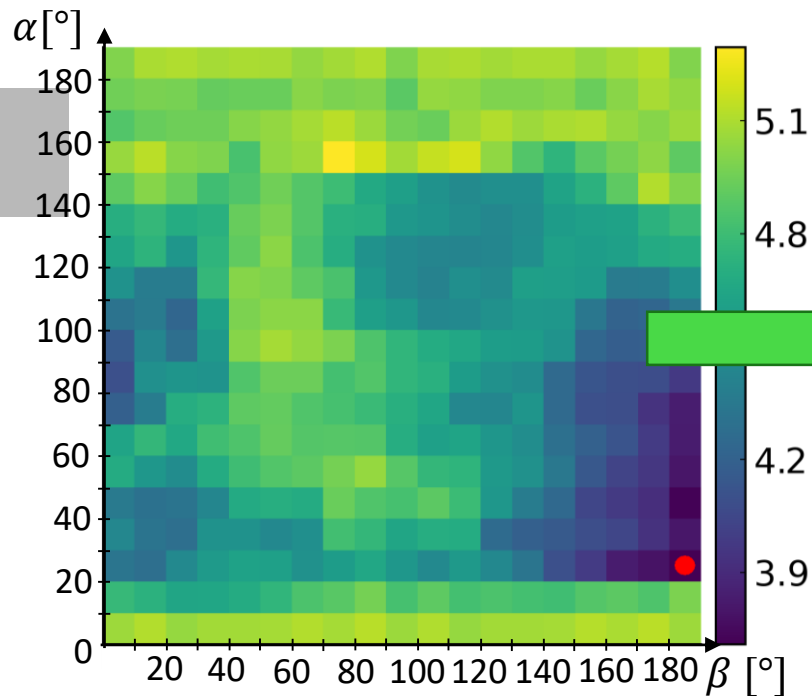
- Models: learn, fail to generalise (overfit)
- No convergence of test losses
- No Model performs better
- Perception of structures
- Distance masks better perceived than in supervised

- Dataset might not be representative (heterogeneous)
-> More data required to compensate
- New framework available for further research !

Solution? Simplify the problem

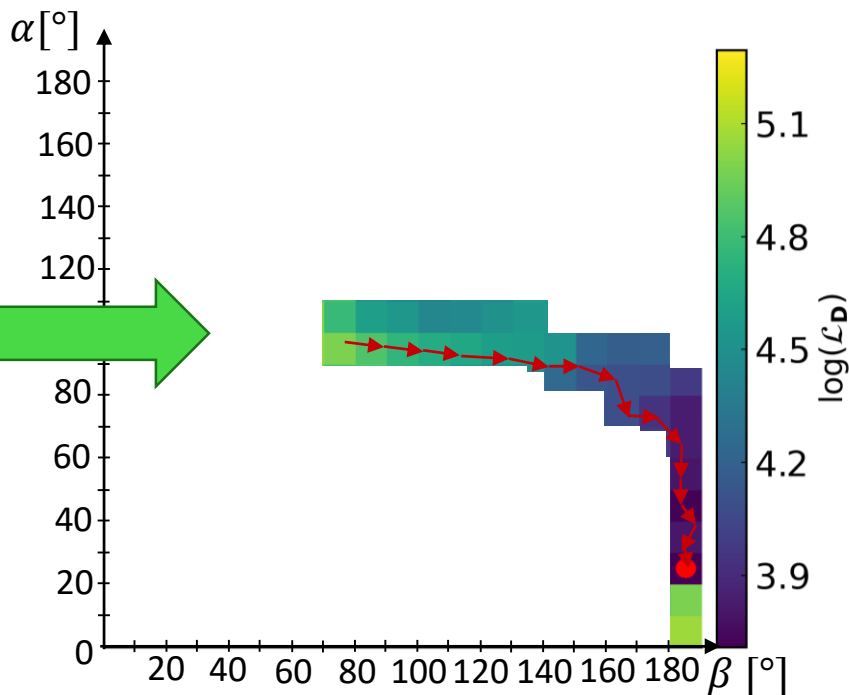
Three Fields Dose Descent

3FDD – Smarter than Brute Force ?



Brute force identification of the minimum for one field.
2h.30min. to global minimum

● Minimum Value



Ideal Gradient Descent for one field.
Reaching global minimum is not guaranteed !

→ Update ● Minimum Value

3FDD – Modified Gradient Descent

$$\vec{\alpha}^{(k+1)} = \vec{\alpha}^{(k)} - \eta_k \mathbf{Grad}_\lambda [\mathcal{L}_D(\vec{\alpha}^{(k)})] \quad 0 \leq k \leq N \in \mathbb{N}$$

where

Prescriptions compliance [7]

- $k \in \mathbb{N} :=$ iteration index
- $\alpha^{(k)} :=$ 3 fields angles – non coplanar !
- $\eta_k :=$ decaying learning rate (schedule)

$$\mathbf{Grad}_\lambda [\mathcal{L}_D(\vec{\alpha}^{(k)})] = \frac{\nabla_{\vec{\alpha}^{(k)}} \mathcal{L}_D(\vec{\alpha}^{(k)}) + \lambda \nabla_{\vec{\alpha}^{(k)}} \mathcal{L}_\pi(\vec{\alpha}^{(k)})}{\|\nabla_{\vec{\alpha}^{(k)}} \mathcal{L}_D(\vec{\alpha}^{(k)}) + \lambda \nabla_{\vec{\alpha}^{(k)}} \mathcal{L}_\pi(\vec{\alpha}^{(k)})\|_2}$$

Physicality of angles

- $\lambda :=$ penalty strength (=1)

3FDD Algorithm :

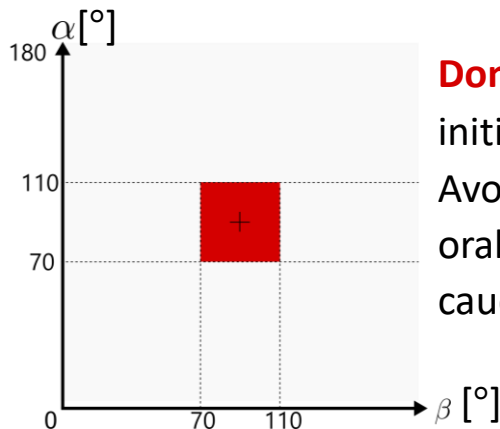
- Initialisation $\vec{\alpha}^{(0)}$
- For $1 \leq k \leq 15$
 - Compute **and store** dose distributions at $\vec{\alpha}^{(k)}$

$$\mathbf{D}_{ijk}(\vec{\alpha}) = \frac{1}{3} \sum_{n=0}^2 \mathbf{D}_{ijk}(\alpha_n, \beta_n) \quad \text{at voxel } v_{ijk}$$

- Compute gradient
- Update angles $\alpha^{(k)}$

Share between fields and iterations -> **speed up !**

Equal field weights !



Domain for random initialisation of $\vec{\alpha}^{(0)}$.
Avoid critical regions :
oral, nasal cavities and
caudal direction

3FDD – Patient and time to solution

Patients Considered

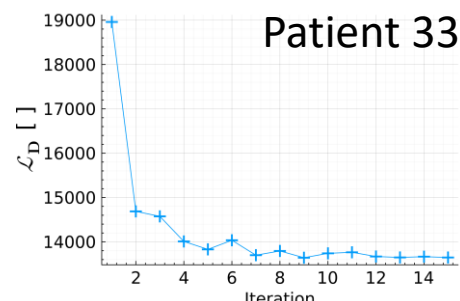
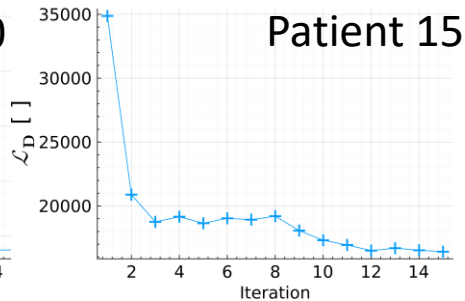
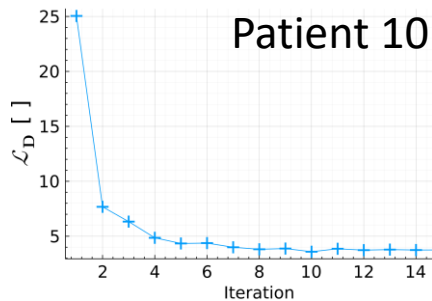
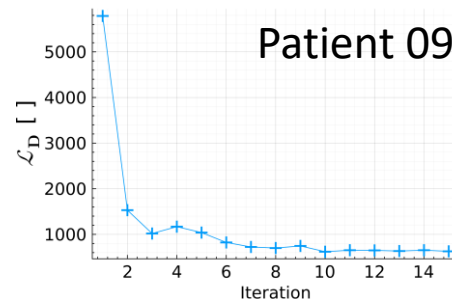
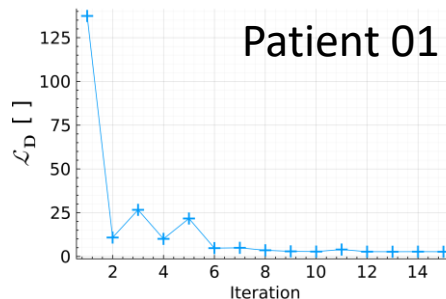
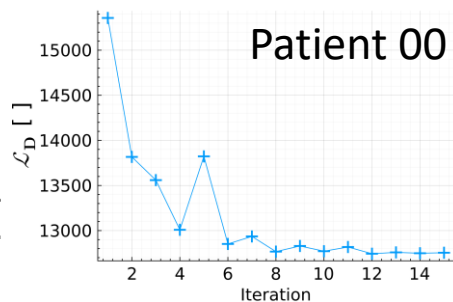
type	ID	Condition	Target Dose [Gy]
Train	00	Chordoma skullbase	74.0
Train	01	Ependymoma	59.4
Train	09	Basalioma (Basal-cell carcinoma)	64.0
Train	10	Glioma low grade	54.0
Train	15	Carcinoma H&N	74.0
Train	33	Meningioma atypical	62.0

Time to solution

Patient Type	ID	Time to solution
Train	0	11 min. 33 sec.
Train	1	08 min. 55 sec.
Train	9	09 min. 20 sec.
Train	10	16 min. 26 sec.
Train	15	17 min. 38 sec.
Train	33	14 min. 39 sec.

Mean Time =
13min±3min40s.

Convergence of the algorithm

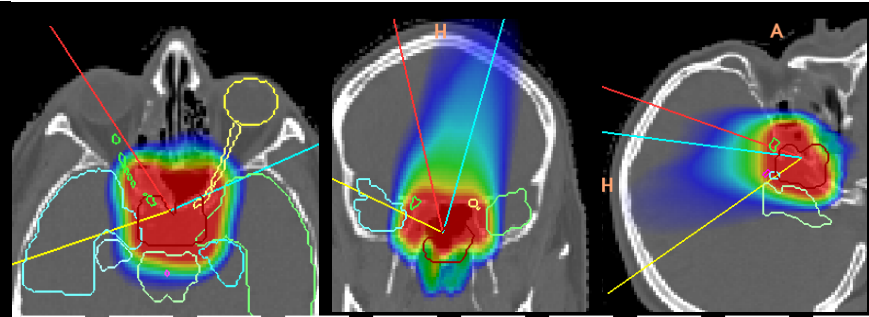
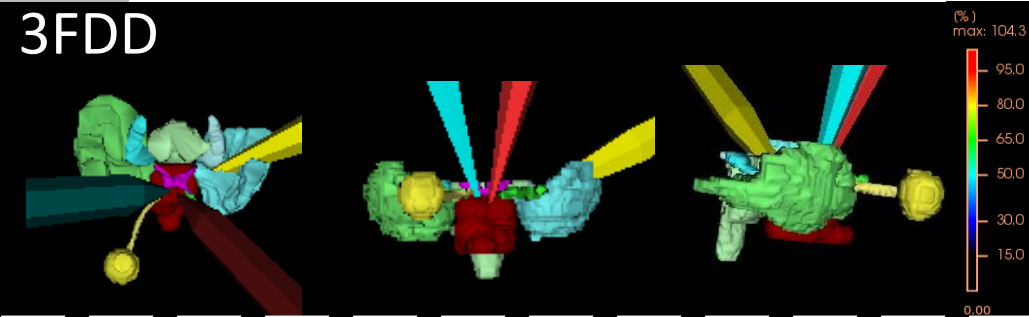


3FDD – Plan comparison patient ID=00

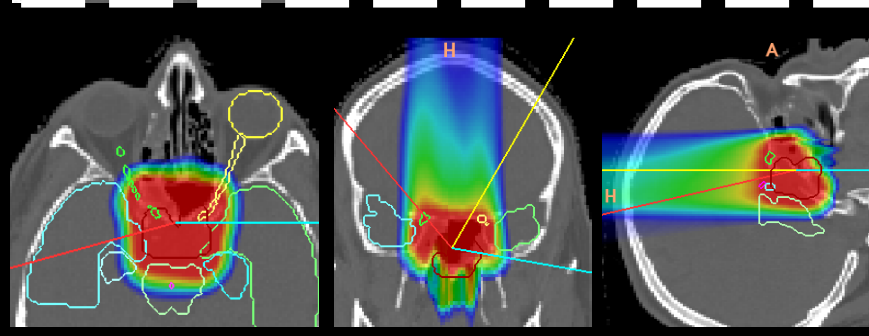
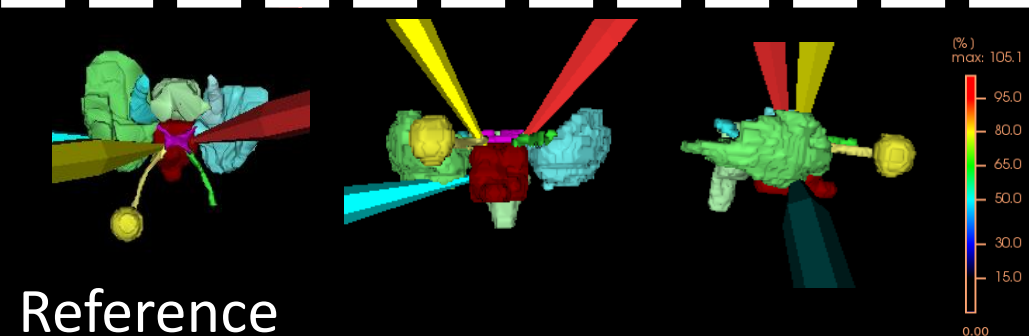
Incidences

Dose distributions

3FDD



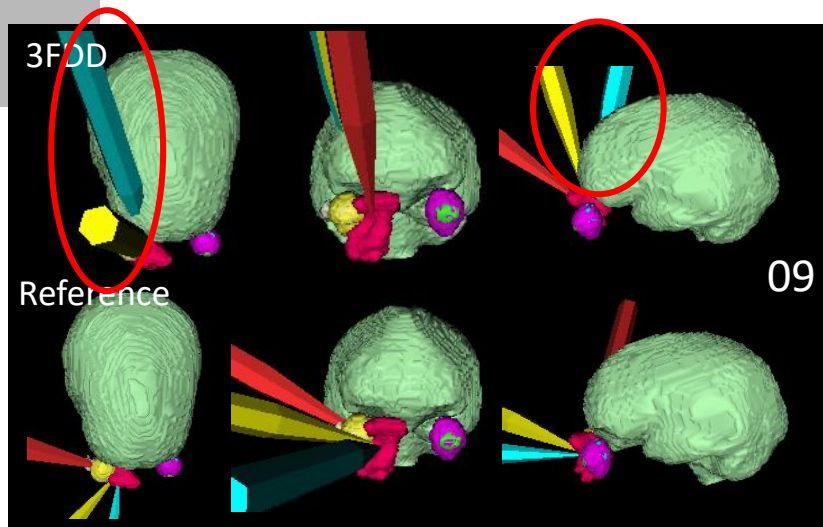
Reference



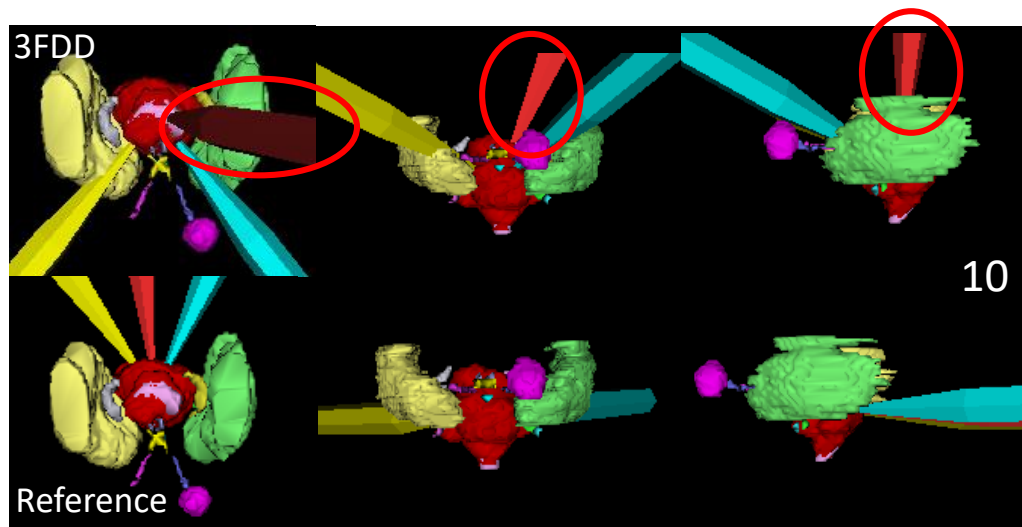
- Comparable performances ✓
- Critical regions avoided ✓

3FDD – Fields Incidences

Problem: Sensitive to patient shifts (high density)



Problem: High path length



- Lacks robustness
- Path length not always minimised

Required Optimisations

1) CNNs

- ❑ Supervised :
 - Median predictor closely beaten
 - Structures perceived
 - No conclusion on augmentation
- ❑ Unsupervised
 - No convergence of test loss
 - Structures perceived
 - Tumour/OAR distinction suboptimal

Outlook

- 1) More homogeneous data required
- 2) Alternative Tumour/OAR distinction
- 3) New Framework for further research

2) 3FDD

Promising results !

- First algorithm for non-coplanar fields with such performances
- Avoidance of forbidden regions
- Competitive time to solution

Outlook

- 1) Robustness to patient shifts
- 2) Path length reduction
- 3) Dose calculation speed
- 4) Fields weights optimisation

My greatest thanks to

- Renato Bellotti
- Dr. Andreas Adelmann
- Prof. Dr. Andreas Pautz

For all the inputs and camaraderie

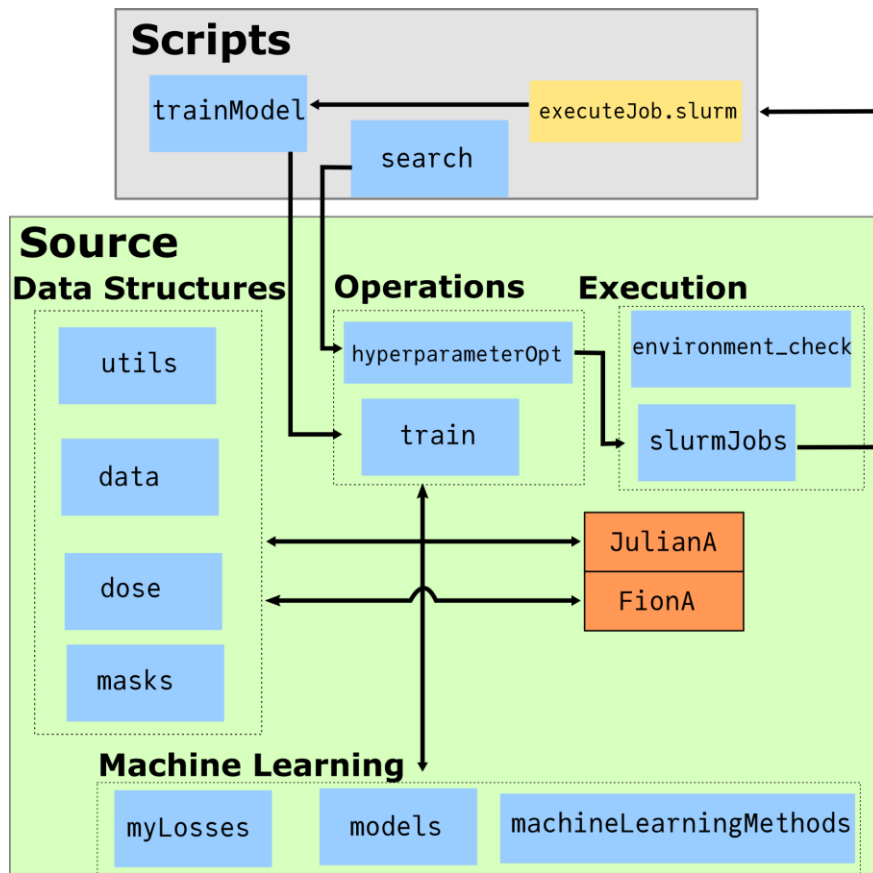
- AMAS team
- WBBB office



Questions ?

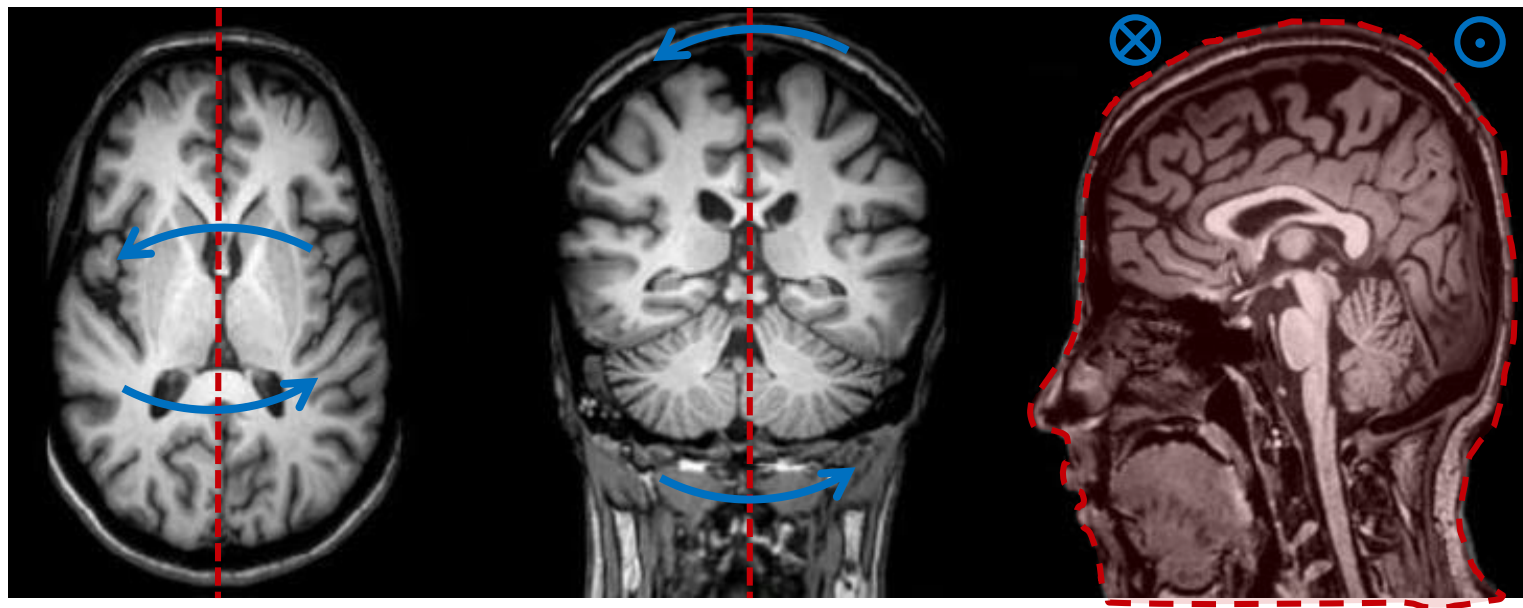
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<https://doi.org/10.1038/sj.bjc.6602754>
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https://www.youtube.com/watch?v=V_6KEDw7xD8
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Appendix - AROBASE



Simplified diagram of AROBASE

Appendix : Expansion – sagittal symmetry

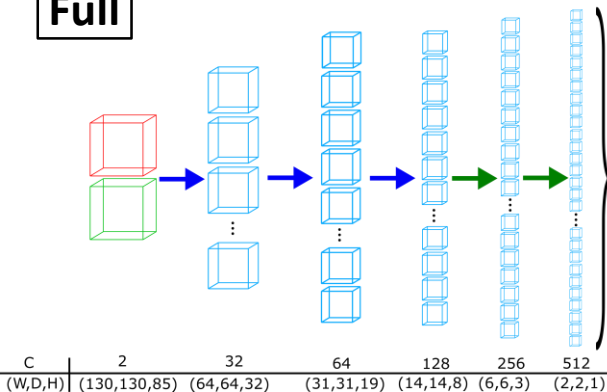


Transformation for patient data expansion using sagittal symmetry of the human head.

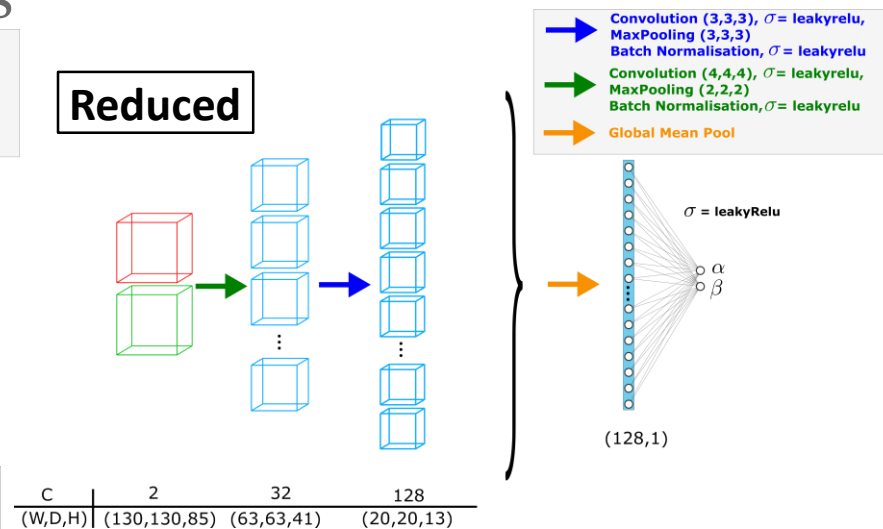
Padmanaban, Sriramakrishnan & Thiruvankadam, Kalaiselvi & T., Padmapriya & Thirumalaiselvi, M. & Sivasakthivel, Ramkumar. (2020). A Role of Medical Imaging Techniques in Human Brain Tumor Treatment. 8. 565-568. 10.35940/ijrte.D1105.12845219.

Appendix - Networks

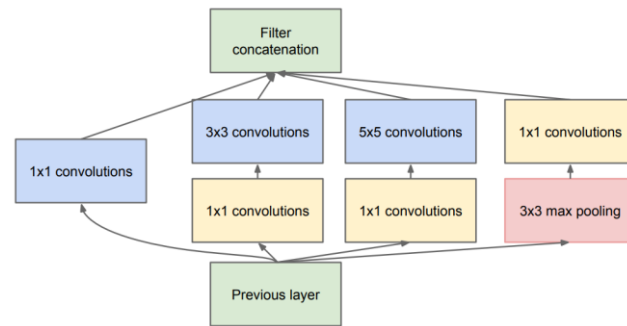
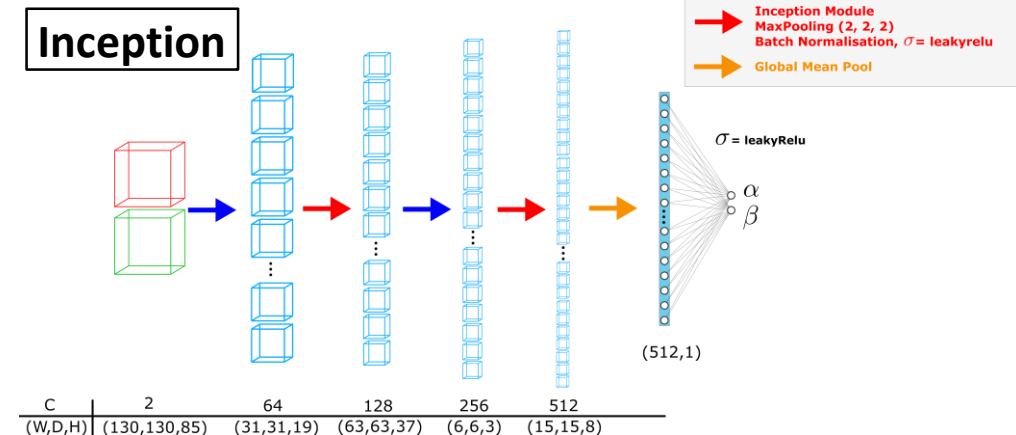
Full



Reduced



Inception



Inception Module [6]

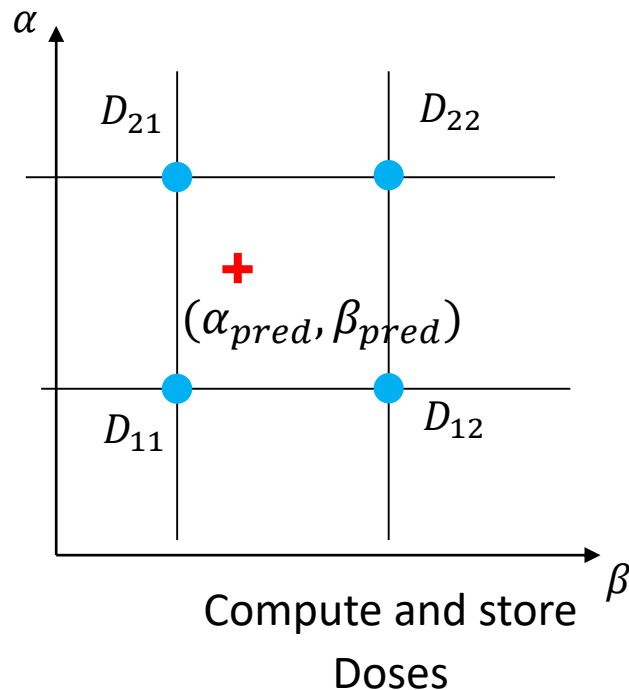
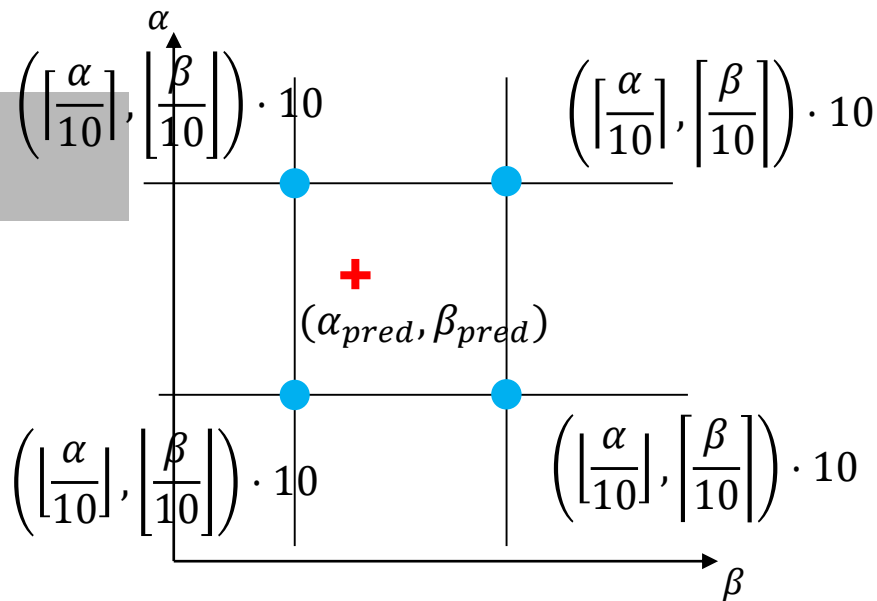
$$L(d(\alpha, \beta)) = \alpha_{ideal} L_{ideal} + \alpha_{hom} L_{hom} + \alpha_{min} L_{min} + \alpha_{max} L_{max} \\ + \sum_o \alpha_{oar_{mean}} L_{mean}(o) + \alpha_{oar_{max}} L_{max}(o)$$

Terms :

- L_{ideal} : aim for ideal dose distribution (from physics of pencil beams)
- L_{hom} : penalizes variance within the targets, aim for a homogeneous dose distribution
- L_{min} : ensure sufficient target coverage (minimum dose distribution)
- L_{max} : all optimization point receive <105% of prescribed dose
- $L_{mean}(o), L_{max}(o)$: mean or maximum dose constraint for OAR

Challenge : penalize out of boundary terms adequately : $L_{penalty} = \exp(\sum_{\alpha, \beta} ||\alpha| - \frac{\pi}{2}|)$

Appendix – Dose calculation (CNN and 3FDD)



$$\mathbf{D}(\alpha, \beta) = \frac{\beta_2 - \beta}{\beta_2 - \beta_1} \left(\frac{\alpha_2 - \alpha}{\alpha_2 - \alpha_1} \mathbf{D}_{11} + \frac{\alpha - \alpha_1}{\alpha_2 - \alpha_1} \mathbf{D}_{21} \right) + \frac{\beta - \beta_1}{\beta_2 - \beta_1} \left(\frac{\alpha_2 - \alpha}{\alpha_2 - \alpha_1} \mathbf{D}_{12} + \frac{\alpha - \alpha_1}{\alpha_2 - \alpha_1} \mathbf{D}_{22} \right)$$

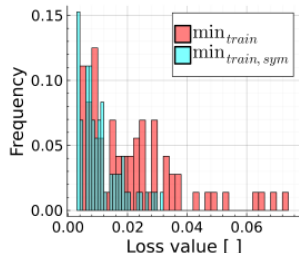
Appendix – Histograms Supervised

Binary Masks

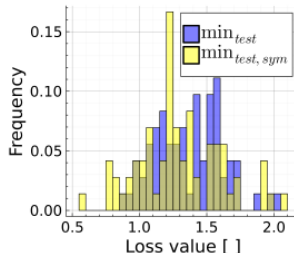
Distance Masks

Histograms of best mean test loss values of each model in the hyperparameter grid search trained in the supervised fashion.

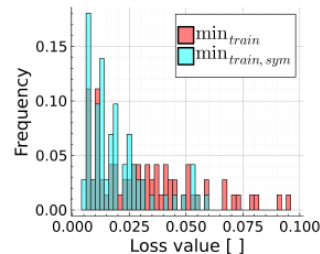
The models were trained with both the initial and augmented dataset (“sym”).



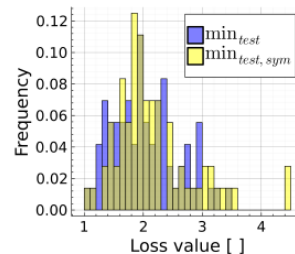
(a) Full Model



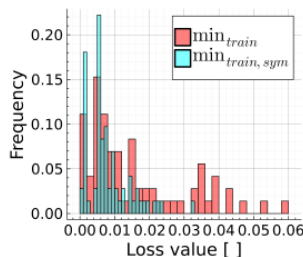
(b) Full Model



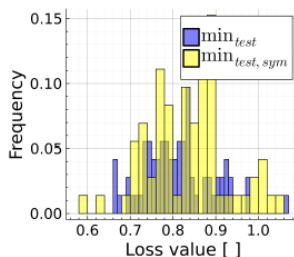
(a) Full Model



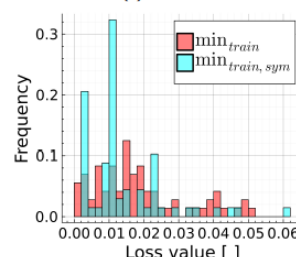
(b) Full Model



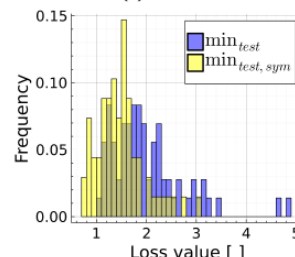
(c) Inception Model



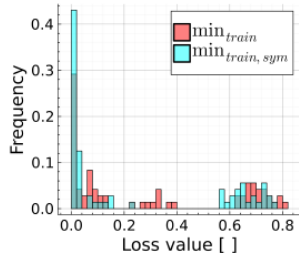
(d) Inception Model



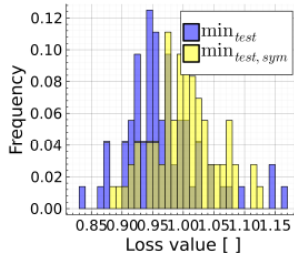
(c) Inception Model



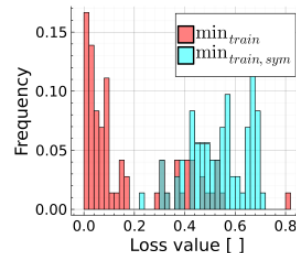
(d) Inception Model



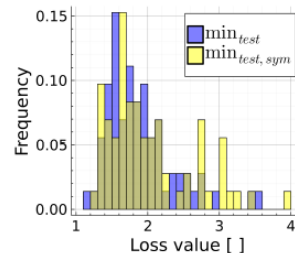
(e) Reduced Model



(f) Reduced Model



(e) Reduced Model



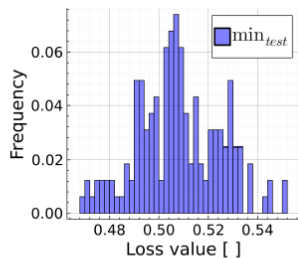
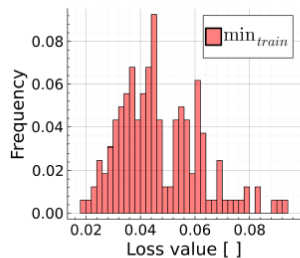
(f) Reduced Model

Appendix – Histograms Unsupervised

Binary Masks

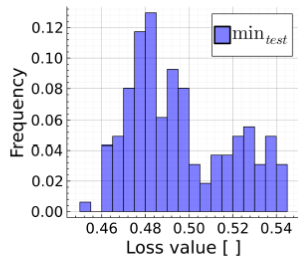
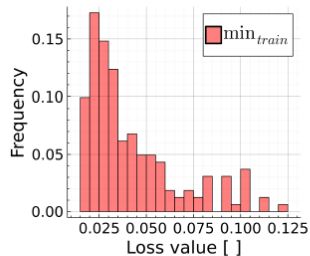
Distance Masks

Histograms of best mean test loss values of each model in the hyperparameter grid search trained in the unsupervised fashion.



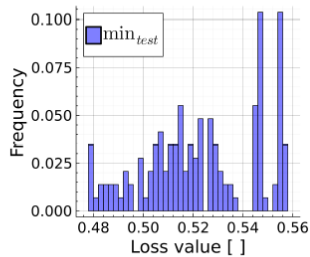
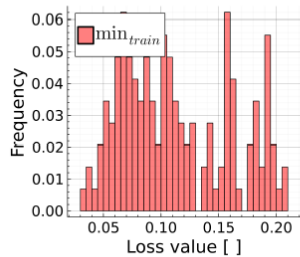
(a) Full

(b) Full



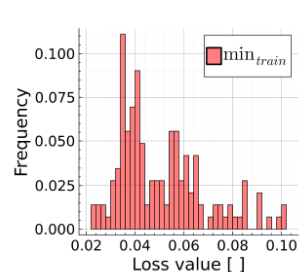
(c) Inception

(d) Inception

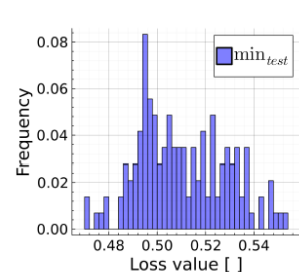


(e) Reduced

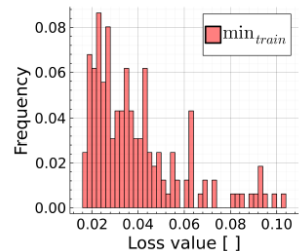
(f) Reduced



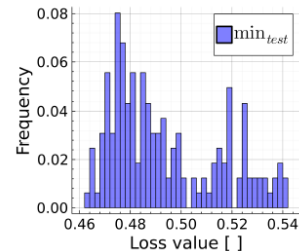
(a) Full



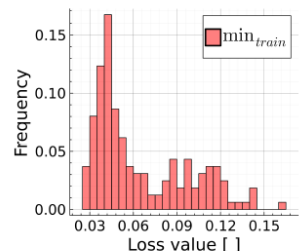
(b) Full



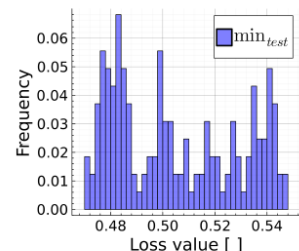
(c) Inception



(d) Inception



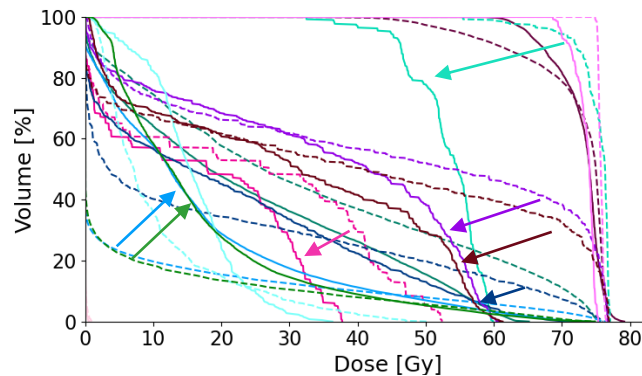
(e) Reduced



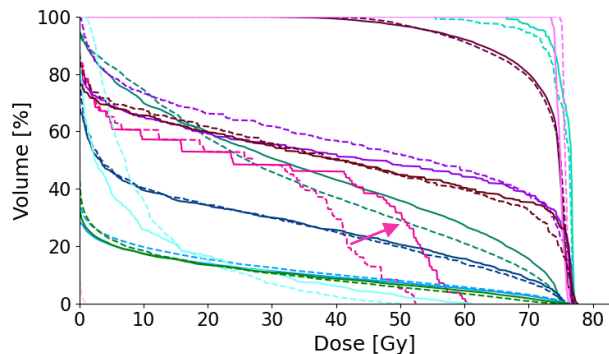
(f) Reduced

Appendix - 3FDD DVHs

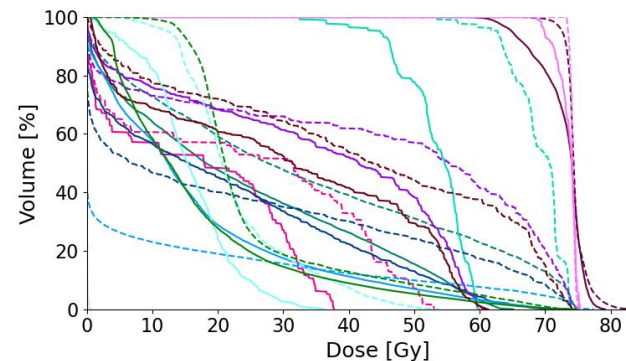
- PTV2 (CTV2+5mm)
- Brainstem
- Center of Brainstem
- Chiasm
- Left Eye
- Left Hippocampus
- Left Optic Nerve
- Left Temporal Lobe
- Pituitary Gland
- Right Hippocampus
- Right Optic Nerve
- Right Temporal Lobe



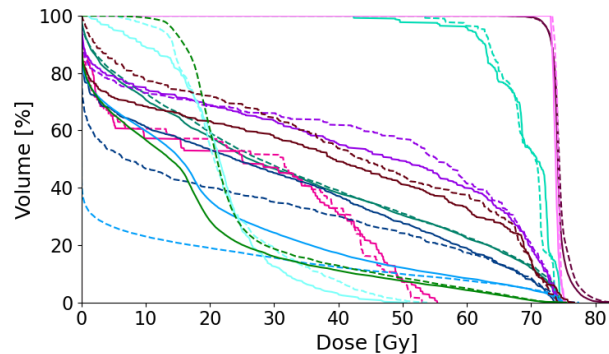
— Clinical Reference
- - 3FDD SFO



— Reference SFO
- - 3FDD SFO



— Clinical Reference
- - 3FDD MFO



— Reference MFO
- - 3FDD MFO

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- [c] Sadeghnejad Barkousaraie, A., Ogunmolu, O., Jiang, S. and Nguyen, D. (2020), A fast deep learning approach for beam orientation optimization for prostate cancer treated with intensity-modulated radiation therapy. *Med. Phys.*, 47: 880-897. <https://doi.org/10.1002/mp.13986>
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1) Radiotherapy methods

- Heuristic score minimisation (Dose objectives) [a]
- Guided pattern search method [b]
- Beam orientation score using CNN (not dose informed) [c]
- Reinforcement learning with MC search tree [d]

2) Proton Therapy methods

- Class Selection [e]
- Beam orientation score using CNN (not dose informed) [f]
- Bayesian 2 step optimization [g]

Appendix - Performances other algorithms

- **Class selection [e]:** 22 minutes (range 9–29 min.). Selection in a list of **predetermined** fields bouquet
- **CNNs as a classifier [f]:** 120h of training, prediction of order seconds. **Coplanar** fields only
- **Bayesian two-steps optimization [g]:** optimisation is performed for two coplanar, two noncoplanar, and three coplanar beams. Iteration optimisation for a given beam configuration takes 2–36 minutes (median times for five patients), acceptable plans are obtained in an average of 23 iterations.